

Education

- 2021–Current **Carnegie Mellon University.**
Ph.D. Student in Human-Computer Interaction
Advisors: [Steven Wu](#), [Kenneth Holstein](#)
Thesis: Principled Measurement and Evaluation of AI Systems under Practical Constraints
Courses: Machine Learning (PhD), Causal Inference, Reinforcement Learning, Behavioral Decision-Making, Modern Techniques in Uncertainty Quantification
- 2020–2021 **Cambridge University.**
MPhil in Advanced Computer Science (Honors with Distinction)
Advisor: [Hatice Gunes](#)
Thesis: Federated Continual Learning for Human-Robot Interaction
- 2015–2019 **University of Missouri.**
B.S. in Computer Science, Psychology (GPA: 4.0/4.0)
Advisors: [Yi Shang](#), [Tim Trull](#), [Steven Hackley](#)

Awards & Honors

- 2023 Best Paper Award at ACM Conference on Fairness, Accountability, and Transparency (FAccT)
- 2023 PhD Fellow, Center for Advancing Safety of Machine Intelligence (CASMI) at Northwestern University
- 2020 National Science Foundation Graduate Research Fellowship
- 2019 University of Missouri Award for Academic Distinction
- 2018 DAAD RISE Fellowship
- 2018 Barry Goldwater Scholarship in Science and Engineering

Academic Publications

[Google Scholar Link](#) (An asterisk (*) below indicates joint first-author.)

Refereed Conference & Journal Publications (Stringently Peer Reviewed)

- 2025 **Luke Guerdan***, Devansh Saxena*, Stevie Chancellor, Zhiwei Steven Wu, and Kenneth Holstein. Measurement as bricolage: How data scientists construct target variables for predictive modeling tasks. In *Proceedings of the 28th ACM SIGCHI Conference on Computer-Supported Cooperative Work and Social Computing (CSCW '25)*, Forthcoming, 2025.
- 2025 **Luke Guerdan**, Solon Barocas, Kenneth Holstein, Hanna Wallach, Zhiwei Steven Wu, and Alexandra Chouldechova. Validating llm-as-a-judge systems under rating indeterminacy. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- 2024 **Luke Guerdan**, Amanda Coston, Kenneth Holstein, and Zhiwei Steven Wu. Predictive performance comparison of decision policies under confounding. In *Proceedings of the International Conference on Machine Learning (ICML '24)*, 2024.
- 2023 Anna Kawakami, **Luke Guerdan**, Yang Cheng, Kate Glazko, Matthew Lee, Scott Carter, Nikos Archiga, Haiyi Zhu, and Kenneth Holstein. Training towards critical use: Learning to situate AI predictions relative to human knowledge. In *ACM Conference on Collective Intelligence (CI '23)*, 2023.

- 2023 **Luke Guerdan** and Hatice Gunes. Federated continual learning for socially assistive robotics. In *Proceedings of the 32nd IEEE International Conference on Robot & Human Interactive Communication (RO-MAN '23)*, 2023.
- 2023 **Luke Guerdan**, Amanda Coston, Zhiwei Steven Wu, and Kenneth Holstein. Ground(less) truth: A causal framework for proxy labels in human-algorithm decision-making. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)*, 2023.
- 2023 **Luke Guerdan**, Amanda Coston, Kenneth Holstein, and Zhiwei Steven Wu. Counterfactual prediction under outcome measurement error. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency (FAccT '23)*, 2023. **Best Paper Award.**
- 2019 Kendall Park, Kourtney Meiss, **Luke Guerdan**, Ev Cheng, Josiah Burchard, John Gillis, Prasad Calyam, and Salman Ahmad. Real-time geotracking and cataloging of mass casualty incident markers in a search and rescue training simulation: Pilot study. In *American Journal of Disaster Medicine*, 2019.
- 2019 Lukas Gehrke*, **Luke Guerdan***, and Klaus Gramann. Extracting motion-related subspaces from eeg in mobile brain/body imaging studies using source power comodulation. In *Proceedings of the IEEE/EMBS Conference on Neural Engineering (NER '19)*, 2019.
- 2018 William Morrison, **Luke Guerdan**, Jayanth Kanugo, Timothy Trull, and Yi Shang. Tigeraware: An innovative mobile survey and sensor data collection and analytics system. In *Proceedings of the IEEE Conference on Data Science in Cyberspace (DSC '18)*, 2018.
- 2017 **Luke Guerdan**, Olivia Apperson, and Prasad Calyam. Augmented resource allocation framework for disaster response coordination in mobile cloud environments. In *Proceedings of the IEEE Conference on Mobile Cloud Computing, Services, and Engineering (MobileCloud '17)*, 2017.
- 2016 Peng Sun, Nicholas Wergeles, Chen Zhang, **Luke Guerdan**, Timothy Trull, and Yi Shang. Ada: automatic detection of alcohol usage for mobile ambulatory assessment. In *Proceedings of the IEEE Conference on Smart Computing (SMARTCOMP '16')*, 2016.

Refereed Workshop & Poster Publications (Lightly Peer Reviewed)

- 2025 Ming Wang, **Luke Guerdan**, Charvi Rastogi, and Kenneth Holstein. Evaluating human-ai decision-making policies under imperfect proxy labels. In *CHI 2025 Workshop on Hybrid Automation Experiences – Communication, Coordination, and Collaboration within Human-AI Teams*, 2025.
- 2023 **Luke Guerdan**, Amanda Coston, Steven Wu, and Kenneth Holstein. Policy comparison under unmeasured confounding. In *NeurIPS 2023 Workshop on Regulatable Machine Learning*, 2023.
- 2022 Anna Kawakami, **Luke Guerdan**, Yang Cheng, Anita Sun, Alison Hu, Kate Glazko, Nikos Arechiga, Matthew Lee, Scott Carter, Haiyi Zhu, and Kenneth Holstein. Towards a learner-centered explainable AI: Lessons from the learning sciences. In *Workshop on Human-Centered Explainable AI (HCXAI) at the ACM CHI Conference on Human Factors in Computing Systems (CHI '22')*, 2022.
- 2022 **Luke Guerdan**, Kenneth Holstein, and Zhiwei Steven Wu. Under-reliance or misalignment? how proxy outcomes limit measurement of appropriate reliance in AI-assisted decision-making. In *Workshop on Trust and Reliance in AI-Human Teams (TRAIT) at the ACM CHI Conference on Human Factors in Computing Systems (CHI '22')*, 2022.
- 2022 **Luke Guerdan**, Amanda Coston, Ken Holstein, and Steven Zhiwei Wu. Ground(less) truth: The problem with proxy outcomes in human-AI decision-making. In *NeurIPS 2022 Workshop on Human-Centered AI (HCAI)*, 2022.
- 2022 **Luke Guerdan**, Amanda Coston, Ken Holstein, and Steven Zhiwei Wu. Counterfactual decision support under treatment-conditional outcome measurement error. In *NeurIPS 2022 Workshop on Causality for Real-world Impact*, 2022.

- 2021 **Luke Guerdan**, Alex Raymond, and Hatice Gunes. Toward affective XAI: facial affect analysis for understanding explainable human-AI interactions. In *Workshop on Responsible Pattern Recognition and Machine Intelligence (RPRMI) at the International Conference on Computer Vision (ICCV '21)*, 2021.
- 2019 **Luke Guerdan***, Peng Sun*, Connor Rowland, Logan Harrison, Zhicheng Tang, Nickolas Wergeles, and Yi Shang. Deep learning vs. classical machine learning: A comparison of methods for fluid intelligence prediction. In *Adolescent Brain Cognitive Development Neurocognitive Prediction: First Challenge, ABCD-NP 2019, Held in Conjunction with MICCAI 2019*. Springer International Publishing, 2019.

Research & Industry Experience

- 2021-Current **Carnegie Mellon University**, *Graduate Student*.
Advisors: [Kenneth Holstein](#), [Steven Wu](#).
Conduct research developing tools for measuring the reliability and validity of algorithmic systems.
- 2024 **Microsoft Research NYC**, *Ph.D. Research Intern*, Fairness, Accountability, Transparency & Ethics (FATE) Team.
Mentors: [Alexandra Chouldechova](#), [Solon Barocas](#), [Hanna Wallach](#).
Developed a statistical framework for principled evaluation of LLMs on subjective NLP tasks.
- 2018-2021 **TigerAware, LLC**, *Co-founder*.
Developed and commercialized mobile-based data collection platform for addiction research. Platform in use by 8,000+ research participants at over a dozen US research institutions, including Brown, Vanderbilt Medical Center, and Massachusetts General Hospital [[web](#), [press](#)].
- 2019 **Washington University in St. Louis**, *Summer Research Intern*, NSF REU in Big Data Analytics.
Advisor: [Chien-Ju Ho](#)
Investigated primal-dual based techniques for efficient online allocation of scarce societal resources.
- 2018 **Technische Universität Berlin**, *Summer Research Intern*, Berlin, Germany.
DAAD RISE Fellowship. Advisors: [Lukas Gehrke](#), [Klaus Gramann](#)
Developed semi-supervised learning algorithms for EEG-based neuroimaging.
- 2017 **ESRI**, *Software Engineering Intern*, Redlands, CA.
Developed geospatial database query optimization algorithms in an enterprise C++/COM stack.
- 2015-2019 **University of Missouri**, *Research Assistant*.
Advisors: [Yi Shang](#), [Tim Trull](#)
Investigated machine learning techniques for understanding human physiology and behavior.
Developed routing algorithms for real-time disaster response coordination (NSF REU in Consumer Networking Technologies, advised by [Prasad Calyam](#)).

Talks & Speaking Engagements

- 2025 **Invited Speaker**. Pareto AI Seminar Series. '*Principled Measurement and Evaluation of AI Systems Under Practical Constraints.*'
Invited Speaker. Amazon Science Responsible AI Seminar. '*Validating LLM-as-a-Judge Systems in the Absence of Gold Labels.*'
Guest Lecturer. Interactive Data Science. '*Principled Evaluation of Human-AI Systems.*'
- 2024 **Invited Speaker**. MILA/McGill NLP Reading Group. Virtual. '*Towards Principled Evaluation Under Imperfect "Ground Truth" Labels.*'

Invited Speaker. INFORMS 24' Session on Human-Centered AI and Decision Making for Social Good. Seattle, WA. '*A Methodological Framework for Human-Algorithm Performance Comparisons Under Incomplete Information.*'

Invited Speaker. EAAMO Meetup at ICML 24'. '*Counterfactual Prediction Under Outcome Measurement Error.*'

Invited Speaker. BIRS Workshop on Bridging Prediction and Intervention Problems in Social Systems. Banff International Research Station, Calgary CA. '*A Framework for Human-Algorithm Performance Comparison Under Incomplete Information.*'

Invited Speaker. Complexity Science Hub Vienna, Lecture Series on Network Inequality. Remote. '*Human-Algorithm Decision-Making Under Imperfect Proxy Labels.*'

2023 **Podcast Guest.** Still Updating Podcast. Remote. '*Finding Truth in Labeled Data.*'

2022 **Invited Speaker** (joint with Anna Kawakami). INFORMS 22' Session on Human-AI Teams. Indianapolis, IN. '*Understanding Human-AI Decision-Making in the Real-World: From Observational Studies to Theoretical Models.*'

2018 **Invited Speaker.** RISE 2018 Intern Summit. Heidelberg, DE. '*Extracting Motion-Related Sub-spaces from EEG in MOBI Studies.*'

Workshops Organized

2024 Jiaee Cheong, Nikhil Churamani, **Luke Guerdan**, Tabitha Edith Lee, Zhao Han, Hatice Gunes, Causal-HRI: Causal Learning for Human-Robot Interaction, HRI.

Grants

2017-2018 **University of Missouri Interdisciplinary Innovations Fund.**
\$25,000. Co-PI with William Morrison.

Service

Reviewing (PC).

2025 NeurIPS, FAccT, ACM Journal on Responsible Computing

2024 ICLR, FAccT, DIS, CHI

2023 ICLR, AAI, CHI, FAccT, ACM Journal on Responsible Computing

2022 CSCW, FAccT

Departmental Service.

2023- Co-organizer of Fairness, Explainability, Accountability, and Transparency (FEAT) reading group.

2021-2023 Graduate Student Assembly (GSA) Departmental Representative.

2021-2023 Carnegie Mellon Graduate Application Support Program Mentor

Leadership.

2015-2019 Mizzou Computing Association. *President, Vice President, Machine Learning SIG Director.*

2017 TigerHacks. *Assistant Director.*

2019-2020 Show Me Dharma Buddhist Society. *Board Member, Director of Communications.*

Mentorship

2025 Kate Lee, Master's Student, Summer Research Assistant.

2023 Ming Wang, High School Student, Summer Research Internship.

2022 Mahika Varma, Undergraduate, HCI Independent Study.

2022 Angelica Bonilla, Undergraduate, CMU AI Mentorship Program.

2021 Matthew Ok, Undergraduate, HCI Independent Study.